

Spacecraft Thruster Firing Test Dataset

Context

Testing hardware to qualify it for Spaceflight is critical to model and verify performances. Hot fire tests (also known as life-tests) are typically run during the qualification campaigns of satellite thrusters, but results remain proprietary data, hence making it difficult for the machine learning community to develop suitable data-driven predictive models. This synthetic dataset was generated partially based on the real-world physics of monopropellant chemical thrusters, to foster the development and benchmarking of new data-driven analytical methods (machine learning, deep-learning, etc.).

Overview of chemical monopropellant thruster

A monopropellant thruster is an engine that provides thrust by using a unique propellant, as opposed to bipropellant systems which use the combustion of fuel and oxidizer. The unique propellant flow into the chamber is controlled by a valve, usually an integral part of the thruster. It is injected into a catalyst bed, where it decomposes. A monopropellant must be a slightly unstable chemical, which will decompose exothermally to produce a hot gas. The resulting hot gases are expelled through a converging/diverging nozzle generating thrust. The gas temperature is high which requires the use of high-temperature alloys to manufacture the nozzle.

The most classical type of monopropellant thrusters are reaction control thrusters generating about 1 to 10 Newton of thrust using hydrazine as propellant. These reaction control thrusters are used, for instance, to control the attitude of a spacecraft and/or to desaturate the reaction wheels.

The performance of a monopropellant thruster (and its degradation) is mostly driven by the valve performance and the state of the catalyst bed on which the propellant decomposes. The life of the catalyst bed is mainly affected by the degradation of catalyst granules. The catalyst is made of alumina-based Indium metal granules (about 1mm in diameter) that are carefully designed and selected to optimize its lifetime. However, catalyst granules are easily damaged by thermoelastic shocks, collisions with other granules, and so on, thus they are broken up into fine particles which reduces their efficiency. After the long duration of firing, large voids are formed in the catalyst bed and induce unstable decomposition of hydrazine and degradation of thruster performance.

The properties of this simulated thruster fire tests are fictitious and not necessarily equivalent to a real-world thruster available on the market. Nevertheless, it provides sufficient granularity and challenge to benchmark an algorithm that may then be tested on real fire test sequences. This is possible because the simulator is based, partially, on real-world physics of such reaction control thrusters. The details of the simulator are not provided on purpose to avoid leakage into feature engineering methods and modelling approaches developed.

Tasks and Use Cases

Supervised Learning Tasks

- **Regression:**
 - **Performance Modelling:** Prediction of the thruster performances (target can be thrust, mass flow rate, and/or the average [specific impulse](#) of over a given sequence that can be calculated using the first two). This task may also be referred to as “time series

forecasting with exogenous inputs”, or “System Dynamics modelling with control”, where the control is the command sent to the thruster (on/off) in the column “ton”. It is very unlikely in practice to have more than a few complete SN qualification datasets, so we suggest a 50:50 split between the train and test sets.

- Use SN01 to SN12 to build your train and validation sets. These are data assumed to be collected on-ground.
- Use SN13 to SN24 as the final test set. Warning: given that these thrusters are expected to be mounted on a flying satellite, you cannot retrain the model progressively using SN13 to SN24 data. The data is the ground truth as if we were able to measure it on-board.
- **Acceptance Test for Individualized Performance Model refinement:**
 - In practice, every newly manufactured thruster is tested via acceptance tests: test_id 1 to 12. So these measurement are actually available for each thruster serial number. For this reason, it is possible to use the acceptance test data (test_id 1 to 12) of SN13 to SN24 as input to the model to predict the flight performance of these SN over test_id 13 to 112. Taking into account the acceptance test of individual thruster might be helpful to generate individualized thruster predictive model. This use case propose to investigate the differences between a one-model-for-all approach or a one-model-for-one-thruster approach.
- **Uncertainty Quantification:**
 - **Thruster-to-thruster reproducibility verification**, i.e. to evaluate the prediction variability between several thrusters in order to construct uncertainty bounds around the prediction (predictive intervals) of the thrust and mass flow rate of future thrusters that may be used during an actual space mission. Use SN01 to SN12 for training and validation, and SN13 to SN24 for test set. The train/val set fraction is lower than in conventional ML settings to mimic real industrial constraints (only a few thrusters fully tested, most of them are flying). The train set is here to generate prediction interval, and the test set (SN13 to SN24) can be used to verify the expected coverage of the predictive interval. As for the regression use case, it may be of interest to include acceptance test data as input data to predict to build and refine predictive intervals and better estimate the per-thruster uncertainties.

Unsupervised / Semi-supervised Learning Task

- **Anomaly Detection:** Anomalies can be detected in an unsupervised setting (outlier detection) or in a semi-supervised setting (novelty detection). The dataset includes a total of 270 anomalies.
 - **Outlier** is an observation which deviates from other observations (abnormal observation). This is an *unsupervised learning approach* where no labels are provided but the algorithm tries to find the datapoints which do not conform to the majority of the observations.
 - **Novelty** is an observation which deviates from known normal observations. This is a *semi-supervised learning approach*: in this scenario the algorithm learns from known nominal data pattern (that have been labelled), or using inliers of an outliers detection system), and flags new observations that deviate significantly from known nominal patterns.
 - There are many ways this dataset can be used to benchmark anomaly detection systems:
 - **Simple approach:** predict if a firing test sequence, given all columns except “vl” and “anomaly code” is anomalous, i.e. the corresponding metadata.csv anomalous flag is

True. A firing sequence is labelled anomalous in the metadata, even if only a few seconds of the time series is anomalous. This a simple yet more practical approach to implement. Traditional metrics such as precision, recall and f1-score can be used to report the anomaly detection performances.

- **Advanced approach:** each firing sequence (time series) comes with an “anomaly_code” column which takes value “0” if it is nominal, or another value (see anomaly codes description) if “anomalous”. It flags for each point in the time series if there is an anomaly or not. It is useful for anomaly detection system that try to predict which portion of a time series is anomalous. Suitable metrics to evaluate the anomaly detection system may be mean intersect over union (IoU) or the Dice Coefficient as the problem is reduced to a segmentation problem where the model is asked to segment the time series into nominal and anomalous portions.

Data Description

The dataset is made of 2,612 csv files, each csv file corresponds to a specific firing test (time series) of different durations. Before diving into the details of these individual csv files, let's have a look at the metadata relative to the dataset. These metadata are crucial to solve the proposed tasks.

metadata.csv

uid	filename	test_id	sn	test_pressure	test_mode	vl1	vl2	vl3	anomalous	anomaly_code	cumulated_throughput	cumulated_on_time	cumulated_pulses
1	00001_001_SN01_24bars_ssf.csv	1	1	24.0	ssf	True	True	False	False	0	0.000000	0.000000	0.0
2	00002_002_SN01_21bars_ssf.csv	2	1	21.0	ssf	True	True	False	False	0	0.451717	0.083333	1.0
3	00003_003_SN01_18bars_ssf.csv	3	1	18.0	ssf	True	True	False	False	0	0.842268	0.166667	2.0
4	00004_004_SN01_15bars_ssf.csv	4	1	15.0	ssf	True	True	False	False	0	1.174725	0.250000	3.0
5	00005_005_SN01_12bars_ssf.csv	5	1	12.0	ssf	True	True	False	False	0	1.452586	0.333333	4.0

Each row contains information about one unique fire test sequence. The columns name description is provided below:

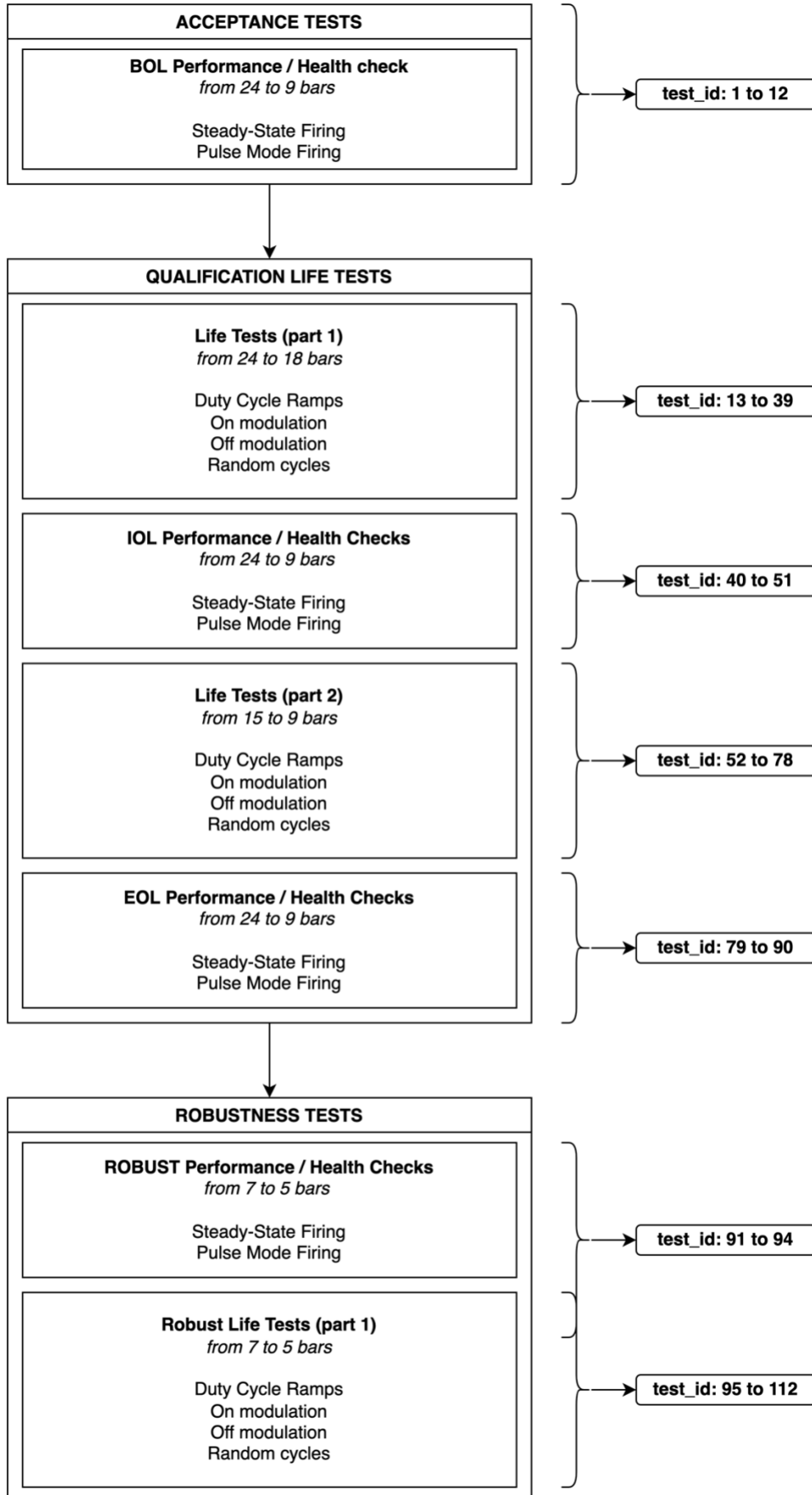
- **uid:** unique identifier of the fire test run. There are chronologically ordered: i.e., uid = 0001 was run prior to uid = 0002, etc.
- **filename:** name of the file containing the detailed sequence test data: command + test measurements
- **test_id:** test identifier: for a given thruster serial number (sn), the test identifier is unique and correspond to a specific, predetermined operating point (same command, same inlet pressure, same test mode).
- **sn:** serial number of the thruster being tested in that sequence. Note that serial numbers reflect the chronological order in which the thruster has been manufactured (and also tested).
- **test_pressure:** pressure at the inlet of the thruster. The higher the pressure, the higher the thrust. The pressure is not expected to vary significantly during the execution of a single firing test.
- **test_mode:** there are different operational mode. Test mode are each imposing different physical constraints on the thruster.
- **vl1:** flag if the series has experienced partial vaporization of the propellant. *This cannot be used as input to a predictive model.*
- **vl2:** flag if the series has experienced vapor lock of level 2 in the injector. *This cannot be used as input to a predictive model.* However it can be used as a target to try to predict which operating conditions will lead to vapor lock level 2.
- **vl3:** flag if the series has experienced vapor lock of level 3, i.e., total blocking of the injector by a propellant bubble. *This cannot be used as input feature to a predictive model.* There should not be any vapor lock level 3 occurrence in the fire test data. If there are, avoidance of such phenomenon is critical and prediction of when this may occur is of key interest to the propulsion teams.

- **anomalous:** target for anomaly detection system: it flags whether an anomaly has occurred or not during the related firing sequence. This does not imply that the entire fire sequence is anomalous. *This cannot be used as input feature to a predictive model.* The anomaly may be
 - only a few milliseconds
 - up to several seconds,
 - or even worst, irreversible (in that only the first sequence in which the anomaly developed itself is marked as anomalous).
- **anomaly_code:** this anomaly code provides supplementary information for evaluation of any anomaly detection system. *This cannot be used as input feature to a predictive model.* It provides a code that refer to a particular kind of anomaly. It is useful to compute statistics about the performance of an anomaly detection system on key slices of the datasets, and to provide insight as to which algorithms works best for a given type of anomaly. The meaning of the anomaly code is provided below.
- **cumulated_throughput:** this is the cumulative mass (in kg) of propellant that has been injected in the thruster and decomposed on the catalyst bed between its beginning of life and until the beginning of the test sequence. This is an ageing factor of the thruster, hence contributing to the degradation of its performances.
- **cumulated_on_time:** this is the cumulative duration (in hours) that the thruster has been “on”, i.e., the valve was commanded open since its beginning of life and until the beginning of the test sequence. This is an ageing factor of the thruster, hence contributing to the degradation of its performances.
- **cumulated_pulses:** this is the cumulative number of pulses (dimensionless) that the thruster has been commanded since its beginning of life and until the beginning of the tests. More precisely this correspond to a counter of the number of times the valve command has been switched from “off” (closed position, no propellant flow) to “on” (open position and propellant flow) since its beginning of life. Every time the valve opens, a fresh amount of propellant is injected on the catalyst bed, leading to a violent and spontaneous exothermic decomposition, thus generating thermo-mechanical shocks that slowly weaken and generate cracks in the catalyst bed. This is an ageing factor of the thruster, hence contributing to the degradation of its performances.

Testing Sequence for each Thruster SN

Before diving into the csv file for each firing sequence, it is relevant to discuss which test sequence is applied to each thruster. It is depicted on the next page.

- **Acceptance Tests:**
 - To asses the performance of newly manufactured units and provide quality control against workmanship defect / material lot deficiencies.
- **Qualification Tests:**
 - To ensure operational regimes and performances are met.
 - To ensure function for the expected mission lifetime.
 - To demonstrate margin to absorb unexpected / unplanned in-flight events or anomalies (usually life test ~ 1.5 x expected mission life)
- **Robustness Tests :**
 - (nice to have) If qualification is successful, robustness tests are there to see how much the unit life can be stretched beyond the minimum required for qualification.



Let's now discuss the individual test sequence data with the collected measurement (thrust and mass flow rate).

Firing Test Sequence CSV files: for instance: "00198_086_SN02_21bars_health_check"

Each firing test sequence csv files follows the naming convention:

"[uid]_[test_id]_SN[sn]_[pressure]bars_[test_mode].csv"

The easiest way to load and handle these files is via the metadata.csv file above-mentioned.

	time	ton	thrust	mfr	vl	anomaly_code
	2022-04-19 06:00:00.000	0	-0.000826	0.000000e+00	0.0	NaN
	2022-04-19 06:00:00.010	0	-0.004935	0.000000e+00	1.0	NaN
	2022-04-19 06:00:00.020	0	0.002713	0.000000e+00	1.0	NaN
	2022-04-19 06:00:00.030	0	-0.000938	0.000000e+00	1.0	NaN

Each firing test sequence correspond to a unique SN (thruster), unique test mode, and unique inlet pressure (between 5 and 24 bars). They include the following column.

- Timestamps of measurement are in the first column
- **ton**: The thruster is operated with a single ON-OFF command. The status of the command is provided in the **ton** column.
- The **thrust** and **mfr** columns are respectively the measured thrust (in Newtons) and measured mass flow rate (in mg/s) recorded at 100 Hz frequency.
- **"vl"**: vapor lock indicator. *This cannot be used as input feature to a predictive model.* See section describing the vapor lock phenomenon.
- **"anomaly_code"**: the target of anomaly detection system, attempt to predict when it is non-zero (therefore anomalous). The actual non-zero values, (anomaly codes) provides supplementary information for evaluation of any anomaly detection system. *This cannot be used as input feature to a predictive model.* It provides a code that refer to a particular kind of anomaly. It is useful to compute statistics about the performance of an anomaly detection system on key slices of the datasets, and to provide insight as to which algorithms works best for a given type of anomaly. The meaning of the anomaly code is provided below.

We assume that SN01 to SN12 have been tested on-ground, therefore all their data are available for training/validation of models.

We assume that SN13 to SN24 are flight units, therefore only the acceptance tests data are available on-ground before the launch. It possible to include these data as input to the model to refine flight predictions as explained in the section regression of Tasks and Use Cases.

Note that anomalies may have occurred in any of the firing sequence, including in the proposed train-validation-test split, and also in acceptance test ids.

Anomaly Codes Description

The following is a list of the anomaly codes. There are useful for the diagnosis of machine learning algorithm performances, for instance to report for which type of anomaly the model makes a lot of errors or perform well on.

The dataset includes a total of 270 anomalies, some more obvious than others. There are all flagged by type with dedicated anomaly codes for diagnosis purpose of the developed machine learning systems.

An anomaly can be temporary or permanent:

- when temporary, this mean it may automatically recover, or be resolved at the end of the test sequence in which it occurred.

- Instead, the consequences of a permanent anomaly will persist on the affected thruster SN in subsequent firing tests. We assume that for anomaly detection we are only interested to detect the firing sequence in which a permanent anomaly materializes but would consider non-anomalous any subsequent firing test sequence despite the consequences still affecting the thruster in the rest of its life. An example of a permanent anomaly is the failure of a component. For example: high thermal shock leading to cracks in the catalyst bed granules, or a solid deposit within the injector tube reducing permanently the mass flow, thus the thrust.

An anomaly can be sudden or progressive:

- Sudden: the amplitude of the effect of the anomaly are immediately visible if the thruster is on.
- Progressive: the effect of the anomaly materializes over several seconds/minutes and may not be immediately visible.

Anomaly Codes

- 1 - mass flow rate leakage / temporary / sudden
- 2 - valve failed closed / temporary / sudden
- 3 - overheating / temporary / sudden
- 4 - response change (type1) / permanent / progressive
- 5 - response change (type1) / permanent / sudden
- 6 - response change (type2) / temporary / progressive
- 7 - response change (type2) / temporary / sudden
- 8 - response change (type2) / temporary / progressive
- 9 - response change (type2) / temporary / sudden
- 10 - slow Isp accelerated degradation / permanent / progressive
- 11 - sudden Isp accelerated degradation / permanent / sudden
- 12 - slow thrust accelerated degradation / permanent / progressive
- 13 - sudden thrust accelerated degradation / permanent / sudden
- 14 - pressure drop / temporary / sudden
- 20 - multi-Spikes anomaly
- 21 - Spike anomaly (type 1)
- 22 - Spike anomaly (type 2)
- 23 - Trend / Drift
- 24 - Some Oscillations (type1)
- 25 - Lots of Oscillations (type2)
- 26 - Signal step
- 27 - Spike anomaly (type 3)
- 28 - Offset anomaly
- 29 - High oscillations
- 30 - Hill anomaly
- 31 - Valley anomaly

These anomaly codes and the dataset has not been provided to develop anomaly classifiers, but to assess the performance of anomaly detection system on different types of anomalies. Therefore, it is considered sufficient to reduce the target to 0 (nominal when anomaly code is 0) and 1 (anomaly code is different from 0), and leverage from the anomaly code only for diagnostic purposes.

Vapor Lock Phenomenon

Vapor lock is a phenomenon that can damage the thruster. Vapor is caused when the propellant flowing through the injector pipe is heated too much (thermal soak back effect generated by the propellant exothermic decomposition) and a fraction of the incoming propellant vaporizes or even worst, creates a bubble that blocks the propellant flow completely. Vapor lock occurs when the chamber is very hot (typically right after long pulses), and when the mass flow rate of fresh, ambient temperature propellant

is not sufficient (short pulses) to compensate for the thermal soak back effect (thermal radiation and conduction within the thruster structure towards the injector). Low inlet pressure, hence low mass flow rate, also tends to favor vapor lock occurrence. Three flags have been created for vapor lock: "vl1", "vl2", and "vl3". There should be no occurrence of "vl3" (extreme vapor lock) in the dataset. One task with the dataset could be to try to predict vapor lock level "vl2", given the commanded ton, inlet pressure, and thruster ageing factor.